Network Services Part – 2

Task 1 → Introduction

The **Network Services 2 room** focuses on identifying and interacting with network services that are commonly found in enterprise environments. These services allow systems to communicate, share resources, and provide functionality such as email delivery, file sharing, and database access.

However, misconfigured services can expose sensitive information and create security vulnerabilities that attackers may exploit.

In this room, we explore several services and learn how to enumerate and interact with them during penetration testing or security assessments.

Learning Objectives

After completing this room, you will understand:

- How to enumerate network services
- How to interact with exposed services
- How misconfigurations can expose sensitive data
- Basic penetration testing techniques for network services

Services covered in this room include:

- NFS (Network File System)
- SMTP (Simple Mail Transfer Protocol)
- MySQL Database Service

Task 2 → NFS (Network File System)

What is NFS?

NFS (Network File System) is a protocol that allows systems to share files over a network. It is commonly used in Linux environments for sharing directories between machines.

Using NFS, a remote system can mount a shared directory and access files as if they were stored locally.

However, if an NFS share is improperly configured, attackers may access sensitive files without authentication.

Enumerating NFS

The first step is scanning the target system to identify open ports.

Example command:

nmap -sV <target-ip>

If **port 2049** is open, it indicates that NFS may be running on the system.

To view exported NFS shares, use:

showmount -e <target-ip>

This command lists all available NFS shares on the server.

Example output:

/home *
/var *

Mounting an NFS Share

Create a directory to mount the share:

mkdir /tmp/mount

Mount the remote directory:

mount <target-ip>:/home /tmp/mount

Now you can view the files:

ls /tmp/mount

Security Risk

Improperly configured NFS shares may allow attackers to:

- Access sensitive files
- Download system configurations
- Retrieve credentials or backup data

Task 3 → SMTP Enumeration

What is SMTP?

SMTP (Simple Mail Transfer Protocol) is used to send email messages between servers.

SMTP typically runs on **port 25**.

Attackers often enumerate SMTP servers to discover valid usernames.

Connecting to SMTP

You can connect to an SMTP server using Telnet:

telnet <target-ip> 25

If the connection is successful, the server will display a banner.

Example:

220 mail.example.com SMTP Service Ready

SMTP Commands

Common SMTP commands include:

Command	Description
HELO	Identify the client
VRFY	Verify if a user exists
EXPN	Expand mailing list
MAIL FROM	Specify sender
RCPT TO	Specify recipient

Example:

VRFY admin

If the user exists, the server will confirm it.

Security Risks

SMTP misconfigurations may allow:

- User enumeration
- Information disclosure
- Email spoofing

Attackers may use this information to perform **password attacks or phishing campaigns**.

Task 4 → MySQL Service

What is MySQL?

MySQL is a widely used relational database management system used by many web applications.

It typically runs on **port 3306**.

Connecting to MySQL

To connect to a MySQL server:

mysql -h <target-ip> -u root

If authentication is weak or misconfigured, attackers may gain access to the database.

Database Enumeration

Once connected, you can enumerate databases.

List databases:

show databases;

Select a database:

use database_name;

Show tables:

show tables;

View table contents:

select * from table_name;

Security Risks

Misconfigured MySQL services may expose:

- User credentials
- Sensitive application data
- Internal system information

Task 5 → Important Enumeration Tools

Nmap

Nmap is used to scan open ports and detect services.

Example:

nmap -sC -sV <target-ip>

Telnet

Telnet allows manual interaction with services.

Example:

telnet <target-ip> 25

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showmount

Used to view NFS shares.

showmount -e <target-ip>

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mount

Used to mount remote NFS directories.

mount <target-ip>:/share /mnt

Task 6 → Importance for Cybersecurity

Understanding network services is essential because attackers frequently target exposed services.

Security professionals use these techniques to:

- Identify exposed services
- Detect misconfigurations
- Perform penetration testing
- Improve network security

These skills are used in:

- Ethical hacking
- SOC analysis
- Incident response
- Red team operations

Conclusion

In this room, we explored how to enumerate and interact with several network services commonly used in enterprise networks.

We learned about:

- NFS file sharing enumeration
- SMTP user enumeration
- MySQL database access
- Network service scanning techniques

Understanding these services helps security professionals detect vulnerabilities and protect network infrastructure from attacks.